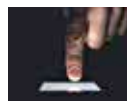




Star Wars Holochess on iPhone

You can now play Star Wars AR Holochess on the iPhone in your pocket with full augmented reality capabilities. <https://dgit.in/arholochess>



Inbuilt smartphone lie detectors

Smartphones could become lie detectors in the future with Veritaps, an algorithm currently in development. <https://dgit.in/veritaps>

I

n today's visual media, visual effects or VFX, designed and developed on computers is as

common as it can get. In fact, one of the major draws of the currently ongoing superhero cinema and TV show wave is the dazzling and copious amounts of computer generated special effects in play. It wasn't always this polished or believable. In the last decade or so, we've seen computer generated imagery, or CGI, improve over time to a point where it seemed like it couldn't get any more real – until it did. But the actual history and inception of CGI goes way beyond that, almost as far back as the dawn of modern computers themselves.

DOTS AND DASHES

One of the earliest notable examples of computer generated animation comes from the Swedish Royal Institute of Technology from way back in 1960. The video (<https://dgit.in/FirstCGI>) shows the render of a car moving at 110 kmph down a highway from the driver's perspective. This particular clip was created on the BESK, or Binär Elektronisk SekvensKalkylator (Binary Electronic Sequence Calculator), Sweden's first electronic computer.

Computer scientists developed more experimental footage in the 1960s. With *Hummingbird* in 1967 and *Kitty* in 1968, the groundwork for physics-based computer powered animations was laid. But the work that would take CGI to a mainstream level was yet to come.

ENTER HOLLYWOOD

In the 1970s, one of the most influential industries in the world, Hollywood, turned its attention towards CGI. Artist and experimental animator Peter Foldes created *Metadata* in 1971. Many claim this to be the first 2D animated feature made entirely on a computer, using the world's first keyframe animation software. Computer animation was mostly inaccessible to artists until this point due to the requisite

LIGHTS, CAMERA, CGI

Computer generated imagery wasn't always as realistic as it is today

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knowledge of computer languages and the mathematical models involved.

The Keyframe Animation system used by Foldes was created by Marcell Wein and Nestor Burtynk at the National Research Council of Canada with the goal of being easy to use for animators. It could be used with a data tablet – akin to drawing on

tablets like an iPad today. The system included a keyboard, a thumb-wheel and a positioning mouse. Perhaps one of the most impressive capabilities in the system for its time was the ability to interpolate between two frames. Checkout its features in more detail here: <https://dgit.in/KFA> and *Metadata* here: <https://dgit.in/Metadata>.



The BESM-4 with an operator



A scene from Metadata



This was the first use of CGI in mainstream



Creepy enough?